APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Sixth Semester B. Tech Degree Regular and Supplementary Examination July 2021

Course Code: EE308
Course Name: ELECTRIC DRIVES

Max. Marks: 100 Duration: 3 Hours

	PART A **Answer all questions, each carries 5 marks.**	Marks
1	**With relevant mathematical derivation, explain the torque equation of motor with rotational motion.**	(5)

A motor generally drives a load (machine) through some transmission system. While motor always rotates, the load may rotate or may undergo a translational motion. Load speed may be different from that of motor, and if the load has many parts, their speeds may be different and while some may rotate, others may go through a translational motion. It is, however, convenient to represent the Torque Equation of Motor Load System by an equivalent rotational system shown in Fig. 2.1

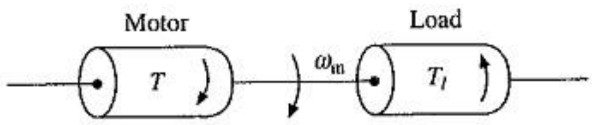

Fig. 2.1 Equivalent motor-load system

Various notations used are:

J = Polar moment of inertia of motor-load system referred to the motor shaft, kg-m^2.

ω_m = Instantaneous angular velocity of motor shaft, rad/sec.

T = Instantaneous value of developed motor torque, N-m.

T_1 = Instantaneous value of load (resisting) torque, referred to motor shaft, N-m.

Load torque includes friction and windage torque of motor.

Torque Equation of Motor Load System of Fig. 2.1 can be described by the following fundamental torque equation:

$$T - T_l = \frac{d}{dt}(J\omega_m) = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt} \qquad (2.1)$$

	Equation (2.1) is applicable to variable inertia drives such as mine winders, reel drives, industrial robots. For drives with constant inertia, (dJ/dt) = 0. Therefore $$T = T_l + J\frac{d\omega_m}{dt} \qquad (2.2)$$ Equation (2.2) shows that torque developed by motor is counter balanced by a load torque T_1 and a dynamic torque $J(d\omega_m/dt)$. Torque component $J(d\omega_m/dt)$ is called the dynamic torque because it is present only during the transient operations. Drive accelerates or decelerates depending on whether T is greater or less than T_1. During acceleration, motor should supply not only the load torque but an additional torque component $J(d\omega_m/dt)$ in order to overcome the drive inertia. In drives with large inertia, such as electric trains, motor torque must exceed the load torque by a large amount in order to get adequate acceleration. In drives requiring fast transient response, motor torque should be maintained at the highest value and Torque Equation of Motor Load System should be designed with a lowest possible inertia. Energy associated with dynamic torque $J(d\omega_m/dt)$ is stored in the form of kinetic energy given by $(J\omega^2_m/2)$. During deceleration, dynamic torque $J(d\omega_m/dt)$ has a negative sign. Therefore, it assists the motor developed torque T and maintains drive motion by extracting energy from stored kinetic energy.	
2	**State and explain how armature current and speed of a dc separately excited motor will be affected by halving armature voltage and field current with load torque remaining constant.** The speed-torque curves for a smooth variation of armature voltage move along the speed axis following changes in the armature voltage. The field winding of the motor is supplied from a separate source. The smooth variation of armature voltage brings about speed control in the zero to base speed range very efficiently. The motor operates in a constant torque mode. This method of controlling the speed of a Separately Excited DC Motor using variable voltage to the armature is employed in Ward Leonard control. The field current is decreased to achieve speeds above base speed when the armature voltage reaches its rated value. The flux weakening mode is best suited for constant power applications, since the armature current may be maintained at its rated value. The torque decreases. In the flux weakening mode the motor cannot be used to drive constant torque loads as the motor draws increased currents as the speed increases. This mode is employed to obtain speeds in the range of base speed to twice base speed. The highest speed attainable by flux weakening is limited by commutation.	(5)
3	**Draw the circuit diagram of a class-C chopper fed DC motor. Draw its V/I characteristics.**	(5)

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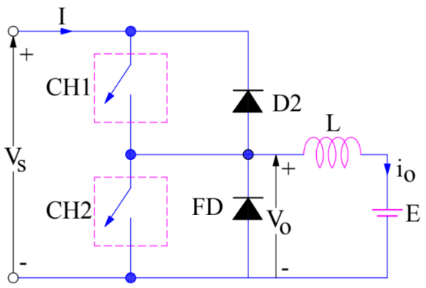

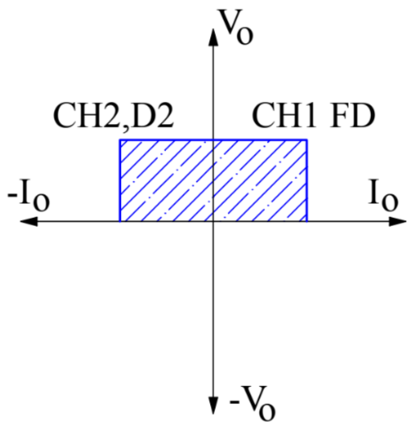

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Explain the closed loop static rotor resistance control method for the speed control of a slip ring induction motor. What are the disadvantages of this method?

A closed-loop speed control scheme with inner current control loop is shown in Fig. Rotor current Ir and therefore, Id has a constant value at the maximum torque point, both during motoring and plugging. If the current limiter is made to saturate at this current, the drive will accelerate and decelerate at the maximum torque, giving very fast transient response. For plugging to occur, arrangement will have to be made for reversal of phase sequence.

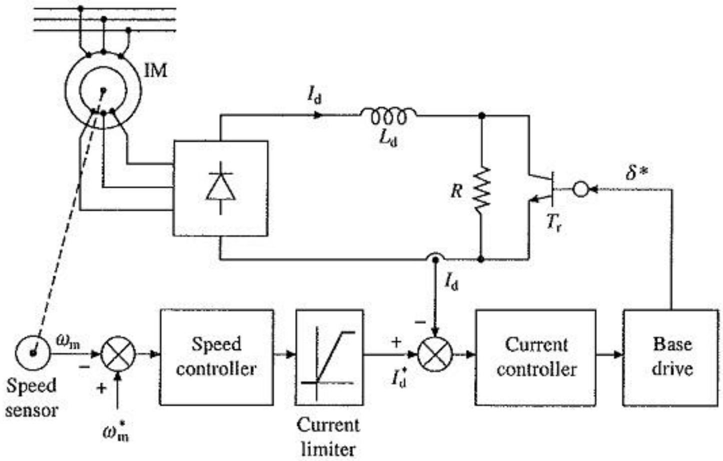

Fig. 6.52 Closed-loop speed control with static rotor resistance control

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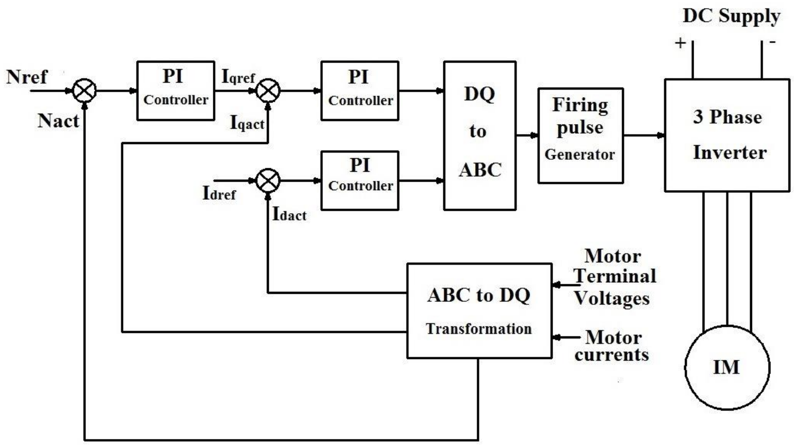
Compared to conventional Rotor Resistance Control of Induction Motor, static rotor resistance control has several advantages such as smooth and stepless control, fast response, less maintenance, compact size, simple closed-loop control and rotor resistance remains balanced between the three phases for all operating points.

5 **With neat block diagram, explain Indirect field-oriented control of IM.** (5)



The direct FOC is problematic and expensive due to use of hall-effect sensors. Hence, indirect FOC methods are gaining considerable interest. The indirect FOC methods are more sensitive to knowledge of the machine parameters but do not require direct sensing of the rotor flux linkages.

The q-axis rotor voltage equation in synchronous frame is

$$0 = r_r i^e_{qr} + \left(\omega_e - \omega_r\right)\lambda^e_{dr} + p\lambda^e_{qr} \qquad (26)$$

Since $\lambda^e_{qr} = 0$ for direct field oriented con

$$0 = r_r i^e_{qr} + \left(\omega_e - \omega_r\right)\lambda^e_{dr} \qquad (27)$$

$$\Rightarrow \omega_e = \omega_r - r_r \frac{i^e_{qr}}{\lambda^e_{dr}}$$

Substituting the values of i^e_{qr} and λ^e_{dr} from **equation 15** and **18** respectively into **equation 27** gives

$$\omega_e = \omega_r + \frac{r_r}{L_{lr}}\frac{i^e_{qs}}{\lambda^e_{ds}} \qquad (28)$$

From **equation 28** it can be observed that instead of establishing θ_e using the rotor flux as shown in **Figure 3** , it can be determined by integrating ω_e given by **equation 28** where ω_e is given as:

$$\omega_e = \omega_r + \frac{r_r}{L_{lr}} \frac{i_{qs}^{e*}}{\lambda_{ds}^{e*}} \qquad (29)$$

The **equation 29** does satisfy the conditions of FOC. In order to check it consider the rotor voltage equations for the q- axis and d- axis:

$$0 = r_r i_{qr}^e + (\omega_e - \omega_r)\lambda_{dr}^e + p\lambda_{qr}^e \qquad (30)$$

$$0 = r_r i_{dr}^e + (\omega_e - \omega_r)\lambda_{qr}^e + p\lambda_{dr}^e \qquad (31)$$

Substituting ω_e from **equation 29** into **equations 30** and **31** gives

$$0 = r_r i_{qr}^e + \frac{r_r}{L_{lr}} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} \lambda_{dr}^e + p\lambda_{qr}^e \qquad (32)$$

$$0 = r_r i_{dr}^e + \frac{r_r}{L_{lr}} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} \lambda_{qr}^e + p\lambda_{dr}^e \qquad (33)$$

Substituting the value of d- axis rotor flux from **equations 17** into **equation 33** gives

$$0 = r_r \left(\frac{\lambda_{qr}^e - L_m i_{qs}^{e*}}{L_{lr}} \right) i_{qr}^e + \frac{r_r}{L_{lr}} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} (\qquad (34)$$

$$0 = r_r i_{dr}^e - \frac{r_r}{L_{lr}} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} \lambda_{qr}^e + p(L_{lr} i_{qr}^e + \qquad (35)$$

If the d-axis rotor current is held constant, then pi^{e*}_{dr}= 0 and rearranging **equations 34** and **35** gives

$$p\lambda_{qr}^e = -\frac{r_r}{L_{lr}} \lambda_{qr}^e - r_r \frac{i_{qs}^{e*}}{i_{ds}^{e*}} i_{dr}^e \qquad (36)$$

$$pi_{dr}^e = -\frac{r_r}{L_{lr}} \lambda_{dr}^e + \frac{r_r}{Lllr} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} \lambda_{dr}^e \qquad (37)$$

6	**What is space vector? What are the applications of reference frame conversion in PWM modulation**	(5)

Space Vector Pulse Width Modulation (SV-PWM) is a *modulation scheme* used to apply a given *voltage vector* to a three-phased electric motor (permanent magnet or induction machine). The goal is to use a steady state DC-voltage and by the means of six switches (e.g. transistors) emulate a three-phased sinusoidal waveform where the frequency and amplitude is adjustable.

There are two challenges to this:

1. The only voltage level available is the DC-link voltage which can be assumed constant (well, at least for sake of simplicity).
2. There are only six different voltage *angles* available. With no middle ground. To rotate a motor, a smoothly rotating voltage vector is required - not one that skips 60 degrees per step.

The significant breakthrough in the analysis of three-phase ac machines was the development of reference frame theory. Using these techniques, it is possible to transform the phase variable machine description to another reference frame. By judicious choice of the reference frame, it proves possible to simplify considerably the complexity of the mathematical machine model. While these techniques were initially developed for the analysis and simulation of ac machines, they are now invaluable tools in the digital control of such machines.

7 **Explain the V/f control characteristics in torque-speed plane of a SM drive** (5)

A drive operating in true Variable Frequency Control of Multiple Synchronous Motors is shown in Fig. 7.9. Frequency command f* is applied to a voltage source inverter through a delay circuit so that rotor speed is able to track the changes in frequency.

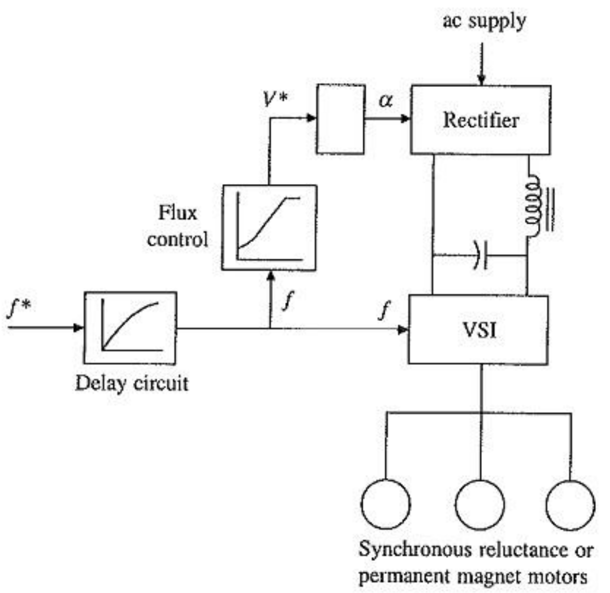


Fig. 7.9 Variable frequency control of multiple synchronous motors

	A flux control block changes stator voltage with frequency to maintain a constant flux below rated speed and a constant terminal voltage above rated speed. This scheme is commonly used for the control of multiple synchronous reluctance or permanent magnet motors in fiber spinning, textile and paper mills where accurate speed tracking between the motors is required. Modes of Variable Frequency Control: Variable Frequency Control of Multiple Synchronous Motors may employ any of the two modes: (i) true synchronous mode or (ii) self-controlled mode, also known as self-synchronous mode. In true synchronous mode, the stator supply frequency is controlled from an independent oscillator. Frequency from its initial to the desired value is changed gradually so that the difference between synchronous speed and rotor speed is always small. This allows rotor speed to track the changes in synchronous speed. When the desired synchronous speed (or frequency) is reached, the rotor pulls into step, after hunting oscillations. Variable Frequency Control of Multiple Synchronous Motors control not only allows the speed control, it can also be used for smooth starting and regenerative braking, as long as it is ensured that the changes in frequency are slow enough for rotor to track changes in synchronous speed. A motor with damper winding is used for pull-in to synchronism. In self-control mode, the stator supply frequency is changed so that synchronous speed is the same as rotor speed. This ensures that rotor runs at synchronous speed for all operating points. Consequently, rotor cannot pull-out of step and hunting oscillations are eliminated. For such applications, the motor may not require a damper winding. In self-control mode, the stator supply frequency is changed in proportion to the rotor speed so that the rotating field produced by the stator always moves at the same speed as the rotor (or rotor field). Since, the voltage induced in the stator phase has a frequency proportional to rotor speed, self-control can be realized by making the stator supply frequency to track the frequency of induced voltage. Alternatively sensors can be mounted on the stator to track the rotor position. These sensors are called rotor position sensors. The frequency of signals generated by these sensors is proportional to rotor speed. Hence, the stator supply frequency can be made to track the frequency of these signals.	
8	**What are the advantages and dis advantages of true synchronous mode of operation?** In true synchronous mode, the stator supply frequency is controlled from an independent oscillator. Frequency from its initial to the desired value is changed gradually so that the difference between synchronous speed and rotor speed is always small. This allows rotor speed to track the changes in synchronous speed. When the desired synchronous speed (or frequency) is reached, the rotor pulls into step, after hunting oscillations. Variable Frequency Control of Multiple Synchronous Motors control not only allows the speed control, it can also be used for smooth starting and regenerative braking, as long as it is ensured that the changes in frequency are slow enough for rotor to track changes in synchronous speed. A motor with damper winding is used for pull-in to synchronism. The motor can also be started by increasing the frequency slowly from zero value. It draws much lower current and produce a much higher torque. As in the case of an induction motor the common control strategy is to operate the motor at a constant air-gap flux upto the base speed and at constant terminal voltage above the base speed. The air-gap flux depends on the value of the magnetizing	(5)

current Im. Hence constant air gap flux operation below base speed is achieved by operating the motor with a constant (v/f) ratio

PART B

Answer any two full questions, each carries 10 marks.

9 (a) Explain the different types of closed loop control configurations of an electric drive. (7)

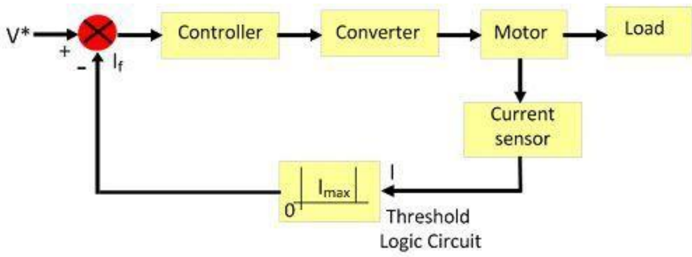
In closed loop system, the output of the system is feedback to the input. The closed loop system controls the electrical drive, and the system is self-adjusted. Feedback loops in an electrical drive may be provided to satisfy the following requirements.

1. Enhancement of speed of torque
2. To improve steady-state accuracy.
3. Protection

The main parts of the closed-loop system are the controller, converter, current limiter, current sensor, etc. The converter converts the variable frequency into fixed frequency and vice-versa. The current limiter limits the current to rise above the maximum set value. The different types of closed loop configuration are explained below.

Current Limit Control

This scheme is used to limit the converter and motor current below a safe limit during the transient operation. The system has a current feedback loop with a threshold logic circuit.
The logic circuit protects the system from a maximum current. If the current is raised above maximum set value due to a transient operation, the feedback circuit becomes active and force the current to remains below the maximum value. When the current become normal, the feedback loop remains inactive.

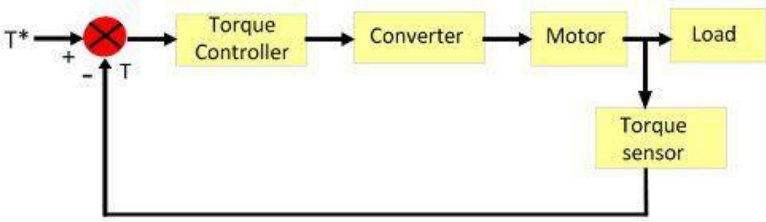


Current Limit Control

Circuit Globe

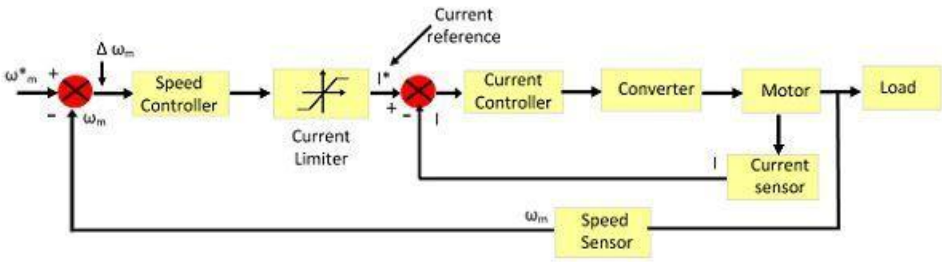
Closed-Loop Torque Control

Such types of loop are used in battery powered vehicles, rails, and electric trains. The reference torque T* is set through the accelerator, and this T* follows by the loop controller and the motor. The speed of the drive is controlled by putting pressure on the accelerator.

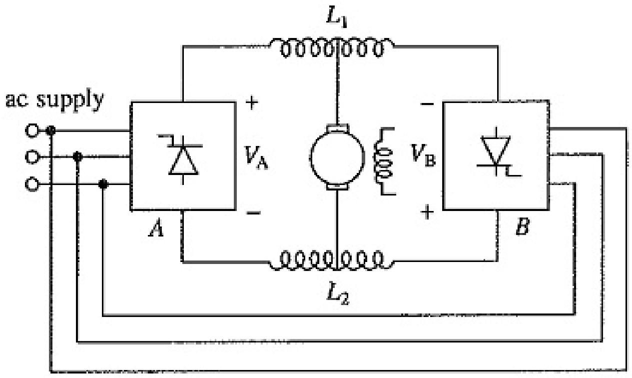
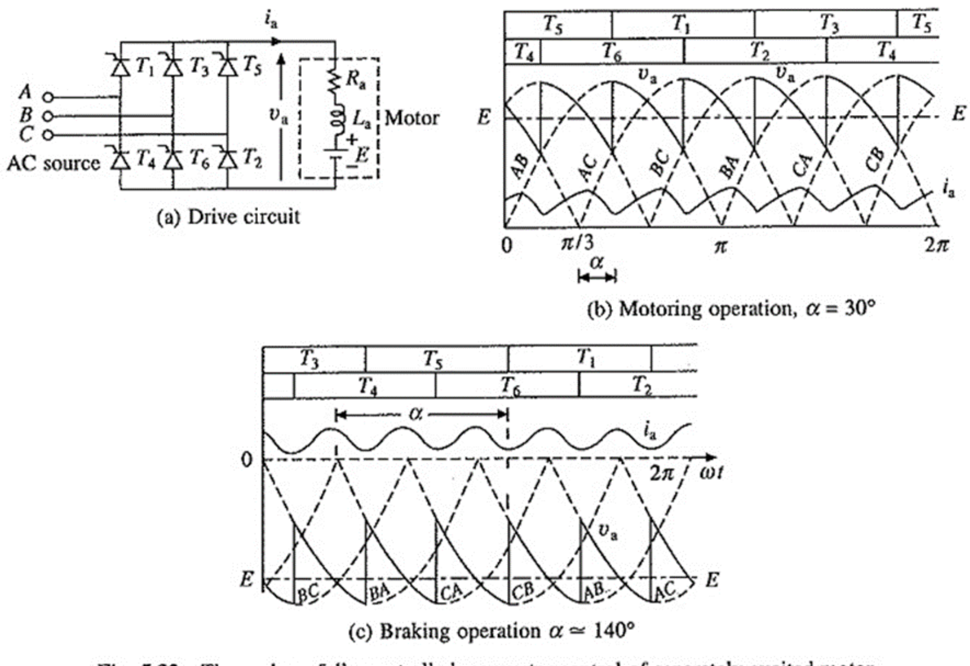


Close Loop Torque Control

Circuit Globe

	Closed-Loop Speed Control The block diagram of the closed loop speed control system is shown in the figure below. This system used an inner control loop within an outer speed loop. The inner control loop controls the motor current and motor torque below a safe limit.  Closed Loop Speed Control Consider a reference speed ω*m which produces a positive error Δ ω*m. The speed error is operated through a speed controller and applied to a current limiter which is overloaded even for a small speed error. The current limiter set current for the inner current control loop. Then, the drive accelerates, and when the speed of the drive is equal to the desired speed, then the motor torque is equal to the load torque. This, decrease the reference speed and produces a negative speed error. When the current limiter saturates, then the drive becomes de-accelerate in a braking mode. When the current limiter becomes desaturated, then the drive is transferred from braking to motoring.	
9 (b)	**What are the factors influencing the choice of an electric drive?** Choice of Electrical Drives depends on a number of factors. Some of the important factors are: Steady state operation requirements: Nature of speed torque characteristics, speed regulation, speed range, efficiency, duty cycle, quadrants of operation, speed fluctuations if any, ratings. Transient operation requirements: Values of acceleration and deceleration, starting, braking and reversing performance. Requirements related to the source: Type of source, and its capacity, magnitude of voltage, voltage fluctuations, power factor, harmonics and their effect on other loads, ability to accept regenerated power. Capital and running cost, maintenance needs, life. Space and weight restrictions if any. Environment and location. Reliability.	(3)
10 (a)	**A drive has the following parameters, J=10 kgm2, T=100-0.1N Nm, active load torque Tl=0.05N Nm, where N is the speed in rpm. Initially the drive is operating in steady state. Later it is reversed and the motor characteristic is changed to T=-100-0.1N Nm. Calculate the time of reversal.** **Solution** Moment of Inertia, J=**10kgm2** Motor Torque, T=**100-0.1N Nm** Load Torque, Tl=**0.05N Nm**	(7)

	The motor is operated at steady state. At steady state T=Tl 100-0.1N=0.05N 100=0.05N+0.1N N=100/0.15=666.67rpm In order to reverse the speed, the motor characteristic is made **= -100-0.1N Nm** Steady state speed after reversing the motor characteristic -100-0.1N=0.05N N=-666.67rpm Ideally infinite time is taken to reach the steady state Therefore, time taken to reach 95% of final state is calculated to find out the time of reversal We know that T=Jdw/dt -100-0.1N=10xd/dt ((2*pi/60)xN) dt=-(10x2xpi)/60{dN/(100+0.1N) on integration with limits of t as 666.67 to (0.95x-666.67) we can reach the answer.	
10 (b)	**How the load torques are classified?** Classification of Load Torques can be broadly classified into two categories- Active and Passive. Load torques which have the potential to drive the motor under equilibrium condition are called Active Load Torques. Such load torques usually retain their sign when the direction of the drive rotation is changed. Torque(s) due to gravitational force, tension, compression and torsion, undergone by an elastic body, come under this category. Load torques which always oppose the motion and change their sign on the reversal of motion are called Passive Load Torques. Such torques are due to friction, windage, cutting etc.	(3)
11 (a)	**Explain the multi quadrant operation of dual converter fed dc motor drives.** The dual converter the name itself means that it has two converters in it. It is an electric device mostly found in variable speed drives. It is a power electronics control system to get either polarity DC from AC rectification by the forward converter and reverse converter. In a dual converter, two converters are connected together back to back. One of the bridge works as a rectifier (converts AC to DC), another bridge works as an inverter (converts DC to AC) and connected commonly to a DC load. Here two conversion processes take place simultaneously, so it is called a dual converter. The dual converter can provide four-quadrant operations.	(5)

	 **Fig. 5.35** Dual converter control of dc separately excited motor. A and B are fully controlled rectifiers. Inductors L_1 and L_2 are used only with simultaneous control Rectifier A which provides positive motor current and voltage in either direction allows motor control in quadrant one and four. Rectifier B provides motor control is quadrants third and four because it gives negative motor current and voltage in either direction.	
11 (b)	**With a neat sketch, explain the motoring and braking operation of three phase fully controlled rectifier control of separately excited DC motor.**  **Fig. 5.32** Three-phase fully-controlled converter control of separately excited motor Three phase Fully Controlled Rectifier Control (6 pulse) fed separately excited dc motor drive is shown in Fig. 5.32(a). Thyristors are fired in the sequence of their numbers with a phase difference of 60° by gate pulses of 120°duration. Each thyristor conducts for 120, and two	(5)

thyristors conduct at a time—one from upper group (odd numbered thyristors) and the other from lower group (even numbered thyristors) applying respective line voltage to the motor. Transfer of current from an outgoing to incoming thyristor can take place when the respective line voltage is of such a polarity that not only if forward biases the incoming thyristor, but also leads to the reverse biasing of the outgoing when incoming turns-on. Thus, firing angle for a thyristor is measured from the instant when the respective line voltage is zero and increasing. For example, the transfer of current from thyristor T5to thyristor T1 can occur as long as the line voltage vAC is positive. Hence, for thyristor T1, firing angle α is measured from the instant vAC = 0 and increases as shown in Figs. 5.32(b) and (c).

If line voltage vAB is taken as the reference voltage, then

Three phase Fully Controlled Rectifier Control

where Vm is the peak of line voltage.

Motor terminal voltage and current waveforms for continuous conduction are shown in Figs. 5.32(b) and (c) for motoring and braking operations, respectively. Devices under conduction are also shown in the figure. The discontinuous conduction is neglected here because it occurs is a narrow region of its operation. For the motor terminal voltage cycle from $\alpha + \pi/3$ to $\alpha + 2\pi/3$ (from Figs. 5.32(b) and (c)).

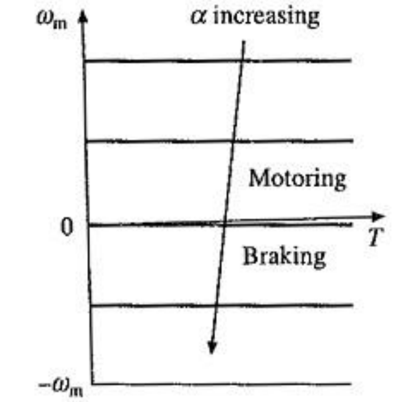


Fig. 5.33 Speed torque curves of drive of Fig. 5.32(a) neglecting discontinuous conduction

12 (a) **With neat circuit and waveforms, explain the midpoint type single phase to single phase cycloconverter.** (5)

A Cycloconverter is a power electronic circuit that converts fixed voltage & frequency AC supply into variable voltage & frequency AC

- Traditionally, AC to AC conversion is done in two different ways

1. In two stages (AC-DC & then DC-AC) as in DC link converters
2. In one stage (AC-AC) as in Cycloconverters

A mid point type cycloconverter configuration is shown in fig.

- There are two groups of SCRs – positive group (T1 & T2) & negative group (T3 &T4)

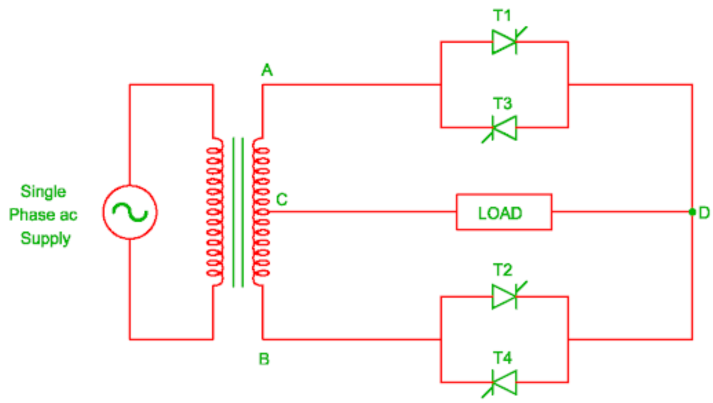


FIG A : SINGLE PHASE TO SINGLE PHASE CYCLOCONVERTER

Step up cycloconverter operation

- SCR T1 is turned ON during positive half cycle of supply at time t=0, therefore load current flows through path A-T1-D-Load-C
- Output voltage is positive during this time
- At t=t1, SCR T1 is force commutated & T4 is turned ON. Now load current flows through path C-Load-D-T4-B
- Out put voltage is negative this time.
- At t=t2, T4 is force commutated & T1 is turned ON
- Now output voltage become positive again
- At t=t3, T1 is force commutated again & T4 is turned On
- As a result output voltage become negative
- At π SCR T2 is turned ON. Now load current flows through B-T2-D- Load-C
- The output voltage become positive
- At t=t4, T2 is turned OFF by force commutation & T3 is turned ON
- The output voltage become negative
- At t=t5, T3 is force commutated & T2 is turned ON
- The output voltage become positive again
- At t=t6, T2 is turned OFF by force commutation & T3 is turned ON
- The output voltage become negative
- Here the output frequency is 4 times input frequency

Step down cycloconverter operation

- During positive half cycle of supply, at t=0 SCR T1 is turned ON
- Now load current flows through A-T1-D-Load-C
- At π, T1 gets naturally commutated and T2 is turned ON
- Now load current flows through B-T2-D-Load-C
- At 2π, T2 gets naturally commutated & T1 is turned ON again
- Now load current flows through A-T1-D-Load-C
- At 3π, T1 gets naturally commutated & T3 is turned ON
- Now load current flows through C-Load-D-T3-A
- At 4π, T3 gets naturally commutated & T4 is turned ON
- Now load current flows through C-Load-D-T4-B
- At 5π, T4 gets naturally commutated & T3 is turned ON
- Now load current flows through C-Load-D-T3-A
- here the output frequency is (1/3) times Input frequency

12 (b) **What are the slip power recovery control schemes used in induction motors? Explain how static Scherbius drive is used to control the speed of induction motors.** (5)

Slip Power Recovery Scheme used in Induction Motor – Figure 6.53 shows an equivalent circuit of a wound-rotor induction motor with voltage Vr injected into its rotor, assuming stator-to-rotor turns ratio unity. When rotor copper loss is neglected

$$P_m = P_g - P_r \qquad (6.92)$$

where Pr is the power absorbed by the source Vr. The magnitude and sign of Pr can be controlled by controlling the magnitude and phase of Vr. When Pr is zero, motor runs on its natural speed torque characteristic. A positive Pr will reduce Pm, and therefore, motor will run at a lower speed for the same torque. When Pr is made equal to Pg, then Pm and consequently speed will be zero. Thus, variation of Pr from 0 to Pg will allow speed control from synchronous to zero speed. Polarity of Vr for this operation is shown in Fig. 6.53 by a continuous line.

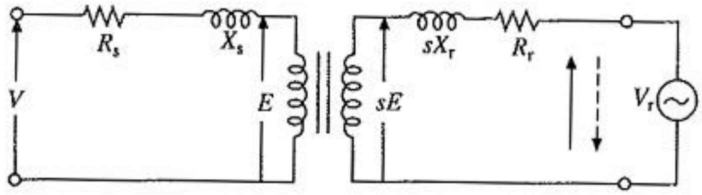


Fig. 6.53 Induction motor operation with an injected voltage in its rotor

When Pr is negative, i.e. Vr acts as a source of power, Pm will be larger than Pg and motor will run at a speed higher than synchronous speed. Polarity of Vr for speed control above synchronous speed is shown by a dotted line in Fig. 6.53.

When rotor copper loss is neglected, Pr is equal to Slip Power Recovery Scheme used in Induction Motor, sPg. Speed control below synchronous speed is obtained by controlling the slip-power. the same approach was adopted in rotor resistance control. However, instead of wasting power in external resistors, it is usefully employed here. Therefore, these methods of speed control are classified as Slip Power Recovery Scheme used in Induction Motor recovery schemes. Two such schemes, Static Sherbius and Static Kramer Drives, are described here.

Static Scherbius Drive :
It provides the speed control of a wound rotor motor below synchronous speed. A portion of rotor ac power is converted into dc by a diode bridge. The controlled rectifier working as an inverter converts it back to ac and feeds it back to the ac source. Power fed back (i.e. Pr) can be controlled by controlling inverter counter emf Vd2, which in turn is controller by controlling the inverter firing angle. The dc link inductor is provided to reduce ripple in dc link current Id.

Since Slip Power Recovery Scheme used in Induction Motor is fed back to the source, unlike rotor resistance control where it is wasted in resistors, drive has a high efficiency. The drive has higher efficiency than stator voltage control by ac voltage controllers because of the same reasons.
Drive input power is the difference between motor input power and the power fed back. Reactive input power is the sum of motor and inverter reactive powers. Therefore, drive has a poor power factor throughout the range of its operation.

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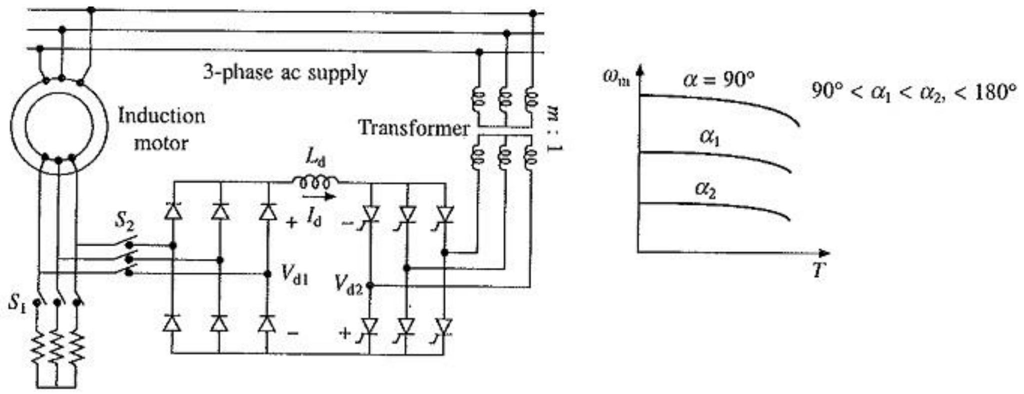
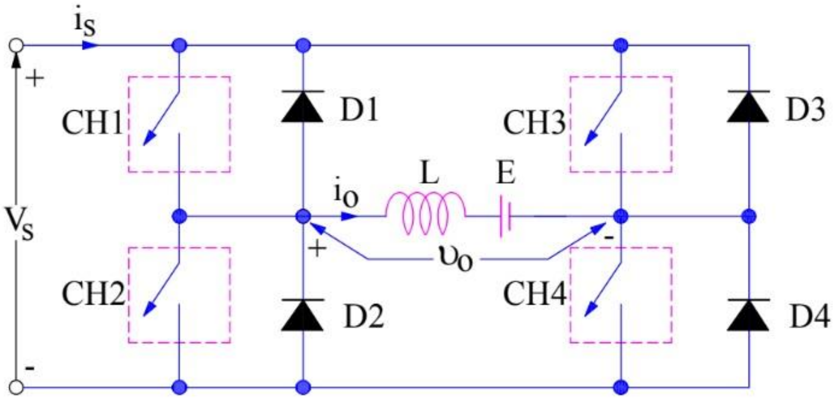
<tr><td></td><td>

(a) The drive circuit (b) Speed-torque curves

Fig. 6.54 Static Scherbius drive</td><td></td></tr>
<tr><td>13 (a)</td><td>

Explain with neat sketch working principle of four quadrant chopper fed DC motor drives. Mention its applications.

A four quadrant chopper is a chopper which can operated in all the four quadrants. The power can flow either from source to load or load to source in this chopper. In first quadrant, a Class-E chopper acts as a Step-down chopper whereas in second quadrant it behaves as a Step-up chopper. This type of chopper is also known as Class-E or Type-E chopper. This article describes the working principle and operation of Class-E chopper with the help of circuit diagram.



Working Principle / Operation of Class-E Chopper:

The circuit of a four quadrant chopper or class-E chopper basically consists of four semiconductor switches CH1 to CH4 and four diodes D1 to D4. The four diodes are connected in anti-parallel. The circuit diagram of this type of chopper is shown below.

In the above circuit diagram, the choppers are numbered CH1 to CH4. For first quadrant operation CH1 is made ON, for second quadrant operation CH2 is made ON and so on. To better understand the working of four quadrant chopper, we will discuss its operation separately for each quadrant.

</td><td>(7)</td></tr>
</table>

First Quadrant Operation:
For first quadrant operation, CH4 is kept ON, CH3 is kept OFF and CH1 is operated. When both CH1 & CH4 are ON simultaneously, the load gets directly connected to the source and hence the output voltage becomes equal to the source voltage. This essentially means that vo = vs. It may be noted that the load current flows from source to load as shown by the direction of io.

When CH1 is switched OFF, the load current free wheels through CH4 and D2. During this period, the load voltage and current remains positive.

Thus, both the output voltage vs and load current io are positive and hence, the operation of chopper is in first quadrant. It may be noted that, Class-E chopper operates as a step-down chopper in this case.

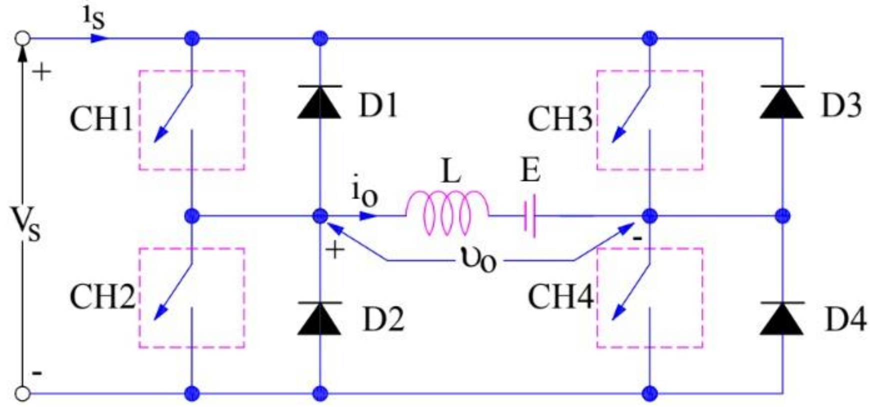
Second Quadrant Operation:
To obtain second quadrant operation, CH2 is operated while keeping the CH1, CH3 & CH4 OFF. When CH2 is ON, the DC source in the load drives current through CH2, D4, E and L. Inductor L stores energy during the On period of CH2.

When CH2 is turned OFF, current is fed back to the source through D1, D4. It should be noted at this point that (E+Ldi/dt) is more than the source voltage Vs. As load voltage Vo is positive and Io is negative, it is second quadrant operation of chopper. Since, the current is fed back to the source, this simply means that load is transferring power to the source. Kindly read Step-up chopper for detailed analysis and better understanding.

For second quadrant operation, load must contain emf E as shown in the circuit diagram. In second quadrant, configuration operates as a step-up chopper.

Third Quadrant Operation:
To obtain third quadrant operation, both the load voltage and load current should be negative. The current and voltage are assumed positive if their direction matches with what shown in the circuit diagram. If the direction is opposite to what shown in the circuit diagram, it is considered negative. One important thing to notice is that the polarity of emf E in load must be reversed to have third quadrant operation. Circuit diagram of Class-E chopper for third quadrant operation is shown below.

For third quadrant operation, CH1 is kept off, CH2 is kept ON and CH3 is operated. When CH3 is ON, load gets connected to source and hence load voltage is equal to source voltage. But carefully observe that the polarity of load voltage vo is opposite to what shown in the circuit diagram. Hence, vo is assumed negative. Let us now see what is the status of load current io. It may be seen that io is flowing in the direction opposite to shown in the circuit diagram and hence negative.

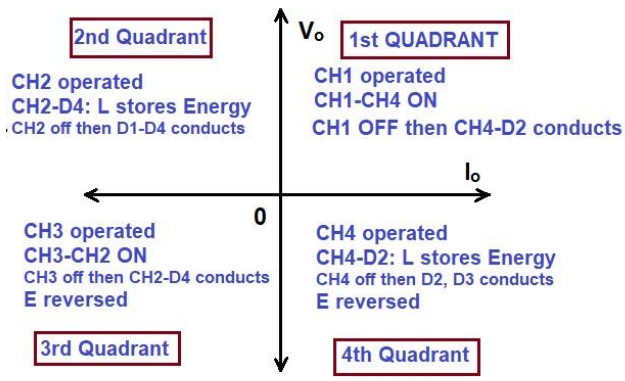


Now, when CH3 is turned OFF, the negative load current free wheels through the CH2 and D4. In this manner, vo and io both are negative. Hence, the chopper operates in third quadrant.

Fourth Quadrant Operation:
To obtain fourth quadrant operation, CH4 is operated while keeping CH1, CH2 and CH3 OFF. The polarity of load emf E needs to be reversed in this case too like third quadrant operation.

When CH4 is turned ON, positive current flows through CH4, D2, L and E. Inductance L stores energy during the time CH4 is ON. When CH4 is made OFF, current is fed back to the source through diodes D2, D3. Here load voltage is negative but the load current is always positive. This leads to chopper operation in fourth quadrant. Here, power is fed back to the source from load and chopper acts as a step-up chopper.

The operation of a four quadrant chopper or Class-E chopper is summarized in the figure below.



13 (b)	**Why chopper-based DC drives give better performance than rectifier-controlled drives.**	(3)

<table>
<tr><td></td><td>The chopper-fed motor is, if anything, rather better than the phase-controlled, because the armature current ripple can be less if a high chopping frequency is used.

Advantages of Chopper Circuits
Chopper circuits have several advantages over phase controlled converters
1. Ripple content in the output is small. Peak/average and rms/average current ratios are small. This improves the commutation and decreases the harmonic heating of the motor.
2. The chopper is supplied from a constant dc voltage using batteries. The problem of power factor does not occur at all. The conventional phase control method suffers from a poor power factor as the angle is delayed.
3. Current drawn by the chopper is smaller than in phase controlled converters.
4. Chopper circuit is simple and can be modified to provide regeneration and the control is also simple.</td><td></td></tr>
<tr><td>14</td><td>A d.c. motor is driven from a class-A d.c. chopper with source voltage of 220 V and at frequency of 1000 Hz. Determine the range of duty cycle to obtain a speed variation from 0 to 2000 rpm while the motor delivered a constant load of 70 Nm. The motor details as follows: 1kW, 200 V, 2000 rpm, 80% efficiency, Ra = 0.1Ω, La= 0.02 H and kϕ=0.54 V/rad /s.
Solution
$$\omega = \frac{2\pi n}{60} = \frac{2\pi \times 2000}{60} = 209.3 \text{ rad/s}$$
$$I_{av} = \frac{P_{out}}{\text{Voltage } \times \eta} = \frac{1000}{200 \times 0.80} = 6.25 \text{ A}$$
$$I_{av} = \frac{\gamma V_d - E_a}{R_a} = \frac{\gamma V_d - K\emptyset\, \omega}{R_a}$$
$$T_{av} = K\emptyset\, I_{av}$$
$$T_{av} = K\emptyset \left(\frac{\gamma V_d - K\emptyset\, \omega}{R_a}\right) \text{Nm}$$
For $\omega_m = 0$
$$T_{av} = \frac{\gamma\, K\emptyset\, V_d}{R_a}$$
$$\gamma = \frac{T_{av}\, R_a}{K\emptyset\, V_d} = \frac{70 \times 0.1}{0.54 \times 220} = 0.058$$
$$\therefore\ \gamma_{min} = 0.058$$
and $$\gamma_{max} = \frac{T_{av} R_a}{K\emptyset V_d} + \frac{K\emptyset \omega_m}{V_d} = \frac{70 \times 0.1}{0.54 \times 220} + \frac{0.54 \times 209.3}{220} = 0.571$$
Hence the range of γ is $0.058 - 0.571$.</td><td>(10)</td></tr>
<tr><td>15
(a)</td><td>Describe the operation of CSI fed induction motor drive. Explain its regenerative braking and multi-quadrant operation.</td><td>(7)</td></tr>
</table>

Current Source Inverter Fed Induction Motor Drive – In a voltage source inverter fed induction motor the applied voltage to the stator is proportional to the frequency, with a correction for the stator resistance drop, particularly at low speeds, to keep the flux constant. It is a very well known fact that the current drawn by an induction motor does not depend on the stator frequency when the air gap flux is constant. There exists a fixed relationship between the slip frequency and stator current for rated flux in the air gap, as shown in Fig. 4.25. By controlling the slip of the motor, the stator current can be controlled. An indirect flux control is therefore possible. The control is simpler than voltage control. The curve between slip frequency and stator current can be calculated using the equivalent circuit. A PWM inverter can be controlled to provide the desired currents in the motor.

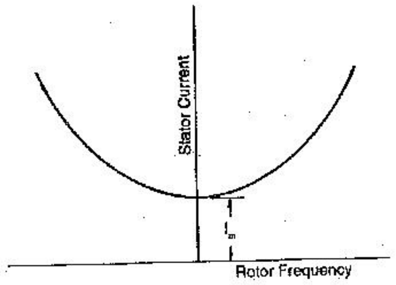


Fig. 4.25 *Stator current of an induction motor as a function of rotor frequency for constant flux*

In a dc link converter, if the dc link current is controlled the inverter is called a current source inverter. The current in the dc link is kept constant by a high inductance and the capacitance of the filter is dispensed with. The variable dc link voltage is converted to a Current Source Inverter Fed Induction Motor by means of the inductance. The dc supply is of high impedance. As the link current is held constant, the output current waveform is determined by the operation of the inverter, while the output voltage is determined by the nature of the load impedance. A Current Source Inverter Fed Induction Motor is suitable for loads which present a low impedance to harmonic currents and have unity p.f.

A Current Source Inverter Fed Induction Motor has a very simple configuration. No feedback diodes are required. A phase controlled rectifier is used on the line side to provide a current control. As the dc link contains only the inductance, regen-eration is possible by changing the polarity of voltages and maintaining the current direction. Hence, a four quadrant drive is simple and straightforward. It provides effective buffering of the inverter output from supply voltage vari-ations. Direct control of stator current allows a precise closed loop control to be implemented with relative ease.

The commutation of the inverter is load dependent. The load parameters form a part of the commutation circuit. A matching is therefore required between the inverter and the motor. Multimotor operation is not possible. The inverter must necessarily be a force commutated one, as the induction motor cannot provide the reactive power for the inverter.

The constant dc link current is allowed to flow through the phases of the motor by control of the inverter, and therefore the motor current is a quasi-square wave. The motor voltage is almost sinusoidal with superimposed spikes, due to commutation. These voltage spikes decide the voltage rating of the thvristors and also affect the insulation of the motor. These spikes can be limited if the machine has small leakage reactance or if the commutating capacitors are large. A machine with smaller leakage reactance is suitable for Current Source Inverter Fed

Induction Motor operation to keep the voltage spikes and the harmonic losses to a minimum. The effect of torque pulsations decreases and the frequency of operation can be increased. The commutating capacitance is chosen to strike a compromise between voltage spikes and the highest operating frequency. The commutation requires a definite minimum current. The inverter has the ability to recover from commutation failure. The link inductance causes a slow rise of the fault current and by the time it reaches high values, the fault may be cleared.

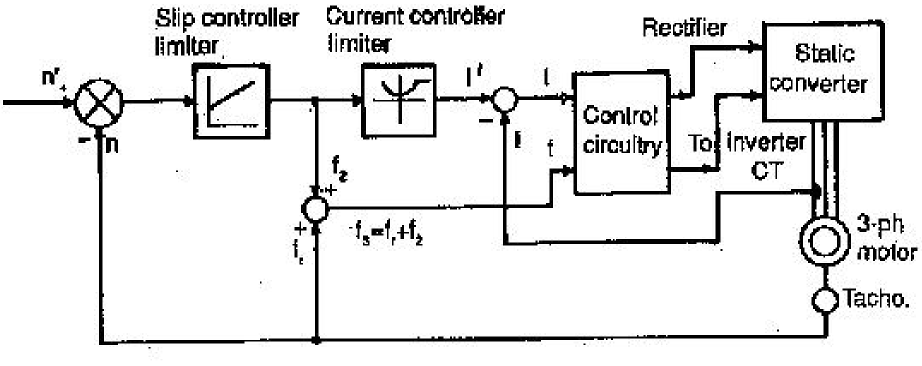


Fig. 4.28 *Slip controlled drive using CSI*

The drive poses stability problems at light loads. Open loop operation is not possible. It has a very wide range of speed control but dynamic performance is poor.
The drive motor requires derating due to harmonic losses and associated heating. Torque pulsations are present and their amplitude is large at low frequency of operation, due to additional harmonics in the rotor flux. The line power factor is poor, due to phase control.

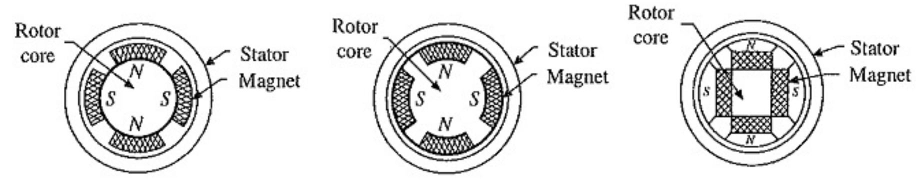
Up to the rated frequency, the drive is in a constant torque mode and above the rated frequency the drive is in a constant horse power mode.
The stator current of an induction motor operating on a variable frequency, variable voltage supply is independent of stator frequency if the airgap flux is maintained constant. However, it is a function of the rotor frequency. The torque developed is also a function of rotor frequency only. Using these fea-tures a slip controlled drive (Fig. 4.28) can be developed employing a current source inverter to feed an induction motor. The relationship between the rotor frequency and stator current for rated flux in the airgap is introduced in the control. Thus an indirect control of flux is possible. The output of the function generator gives the reference value of current. The measured current is compared with the reference value and the error is used to alter the firing angle of the phase controlled converter on the line side. The in-put to the function generator is the difference between the reference speed and actual speed and it can be considered as slip frequency which is added to the frequency corresponding to rotor speed This gives the value of stator frequency and the machine side inverter is controlled to give this frequency. The control is operative until the rotor attains the desired speed with the required slip frequency.

15 (b) **Explain in detail about the classification of PM synchronous motor?** (3)

Permanent Magnet Synchronous Motors Types are classified as:

surface mounted and interior (or buried).
Surface mounted PM motors are of two types: (1) projecting type, in which magnets project from the surface of rotor (Fig. 7.1(a)), and (2) inset type, in which magnets are inserted into

the rotor, providing a smooth rotor surface (Fig. 7.1(b)). Epoxy glue is used to fix the magnets to the rotor surface in both. While these motors are easy to construct and are less expensive, they are less robust compared to interior type (Fig. 7.1(c)) rotors and are not suitable for high speed applications. In interior type PM motors, magnets are imbedded in the interior of the rotor.



(a) Projecting surface mounting (b) Inset magnet (a) Interior magnet

Fig. 7.1 Different types of PM synchronous motors

The wound field and permanent magnet Synchronous Motors Types have a higher full load efficiency and power factor than an induction motor. Wound field motors can be designed for a higher power rating than induction motors. Since the air-gap flux is not produced solely by the magnetizing current drawn from the armature, a larger air-gap suiting the mechanical design can be chosen. The ability to control power factor is an important advantage at higher power levels. Operating at unity power factor minimizes the inverter rating. Apart from the robust construction, permanent magnet synchronous motor has low losses and high efficiency. Because of low losses, it is possible to make motors with very high power density and torque to inertia ratios. These make them suitable for servo drives requiring fastest possible dynamic response.

The rotor of a synchronous reluctance motor has salient poles but neither have field winding nor permanent magnets. Motor is driven by reluctance torque which is produced due to tendency of the salient rotor poles to align themselves with synchronously rotating field produced by the stator.

All the above mentioned Synchronous Motors Types, when designed to operate with a source of fixed frequency, are provided with damper winding, which is similar to squirrel-cage winding of an induction motor. It is used to start the machine as induction motor and to damp the hunting oscillations which occur during the transient operations. When fed from a variable frequency source, capable of smooth frequency variation from zero to rated, the damper winding is not required for starting. It may, however, be required for damping hunting oscillations or for some other purposes explained later.

One important difference between the wound field and permanent magnet motors, which are designed to operate with a source of fixed frequency, must be noted. When a wound field motor is started as an induction motor, dc field is kept off. In case of a permanent magnet motor, the field cannot be ‘turned off’. When at a speed below synchronous speed, the rotor field induces a voltage in the stator, which has a frequency different than the frequency of stator supply. The current produced by induced voltage interacts with the rotor field to produce a braking torque, which opposes induction motor torque due to damper winding. The permanent magnet synchronous motor (PMSM) is designed so that the braking torque is very small compared to induction motor torque. Because of the capability to start direct on line these motors are called line start PMSM. They are available in 3-phase and 1-phase construction. Although expensive compared to induction motors, they have advantages of high efficiency, high power factor and low sensitivity to supply voltage variations. Therefore, they are preferred for industrial applications with large duty cycles such as pumps, fans and compressors.

The hysteresis synchronous motors are employed in low power applications requiring smooth start and quiet operation.

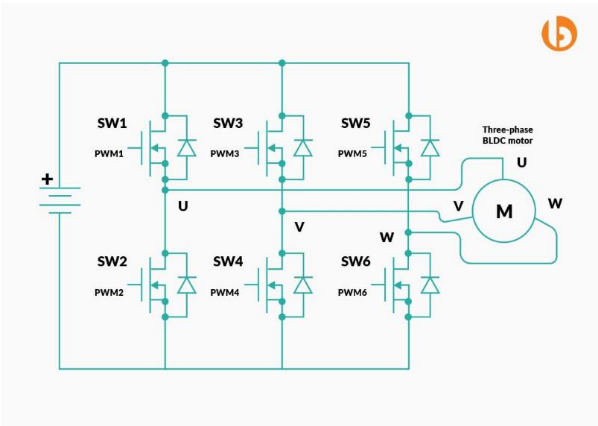
16 (a) **If the induced emf in the stator of an 8-pole induction motor has a frequency of 50 Hz and that in the rotor is 1.5 Hz, at what speed is the motor running and what is the slip?** (4)

Slon: f = 50 Hz and fr = 1.5 Hz,
Ns = 120f/p = 120 x 50/8 = 750 rpm
We have fr = sf
Hence s= fr/f = 1.5/50 = 0.03

16 (b) **Briefly describe the working of trapezoidal permanent magnet AC motor.** (6)

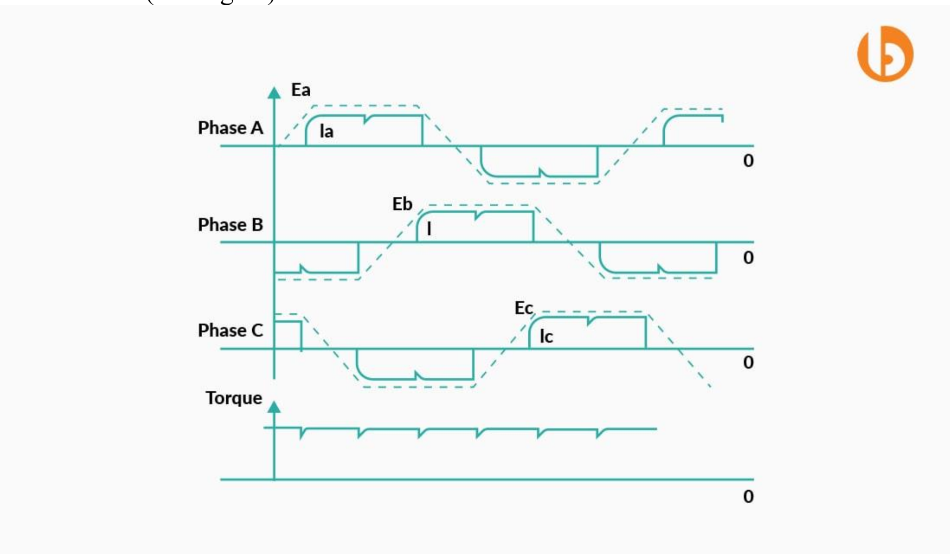
With the Brushless motor technology, you can get high reliability along with efficiency in your applications; that too at a lower cost as compared to the brushed motors. Brushless DC motor and Permanent magnet synchronous motor(PMSM) both consist brushless characteristics.Both motors are synchronous machines.There’s very small difference between them.The difference is the shape of their back emf.BLDC and PMSM motor has respectively trapezoidal and sinusoidal back emf.In reality motor can not produce exactly trapezoidal back emf but it is more like to sinusoidal. So sometimes type of motor doesn’t matter but the commutation(control) method matters more than that.

It is necessary to have stator winding currents have the same shape as its back EMF to get maximum efficiency.The drive current is given to three phase motor from the inverter depending upon PWM applied to it.PWM is generated using the controller.Stator winding current is changing according to the rotor position.Rotor position is acquired by the HALL sensor,QEP encoder or any other mathematical methods.The different method is used for the drive current generation depending upon the back EMF of the motor.



Trapezoidal Control

Trapezoidal commutation method defines the shape of back EMF and drive current waveform. To get the optimum performance from the motor, the drive current should match the back Electro Motive Force waveform. As back EMF of BLDC motor is in trapezoidal shape it should be driven with trapezoidal current to get best performance. Trapezoidal commutation is also known as “Six Step Commutation” because there are total six steps of drive current used to complete one revolution of rotor. In trapezoidal control at a time only two phase are active. This method has simple control algorithm but the torque ripple exist in the motor at every commutation(60 Degree).



Advantages

Control Algorithm is simple

Only two phase is needed to active at time

Less switching losses

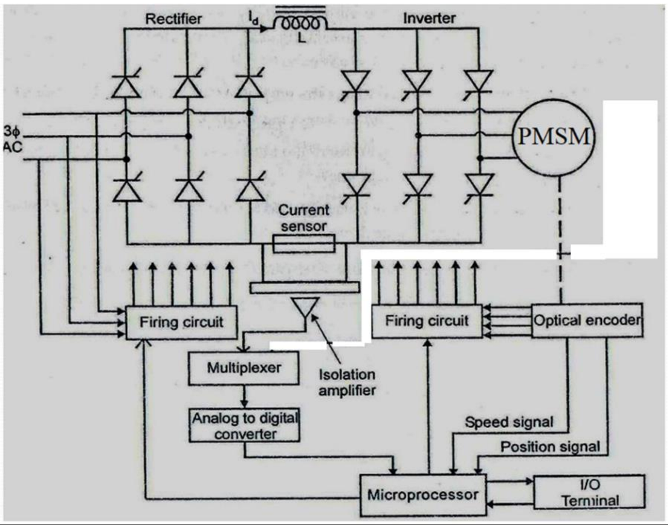
Disadvantages

Torque ripple at every commutation

Torque produced is less

Acoustic and electric noise

17 (a) **Draw the block diagram of microcontroller-based control of permanent magnet synchronous motor drive.** (6)



17 (b) **Explain the modes of operation of PMSM drives.** (4)

Figure 1 shows the speed and torque variation of the PMSM machine in a different mode of operation namely forward motoring (FM), forward regeneration (FR), reverse motoring (RM) and the reverse regeneration (RR) mode. The fist quadrant is the FM mode in this motor rotates in forward direction as the speed and torque is positive, the third quadrant is the mirror reflection of the first quadrant here speed and torque both are in negative direction. The negative torque opposes the positive motoring torque, due to this power flow from the machine to the power supply source this mode is called as the reverse braking mode. Similarly the quadrant III and II, the difference between the I and IV quadrant is the direction of rotation, hence the phase sequence is changed.

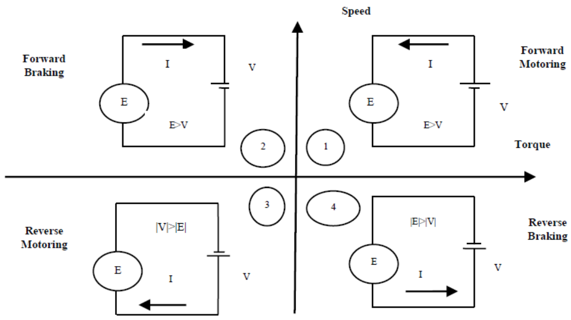


Figure 1: Four Quadrant operation of PMSM

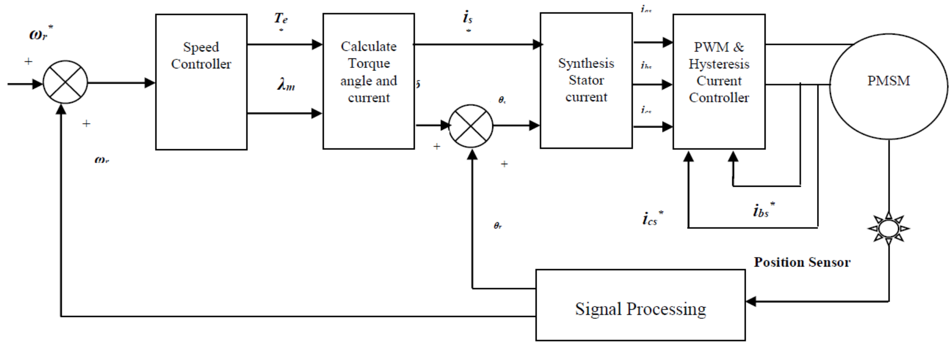


Figure 2: Speed Controlled PMSM Drive

A speed-controlled PMSM drive system having a two loop, the inner loop is for the torque controlled that is the basic core of the drive system and an outer loop is the speed control loop for controlling the rotor speed of the drive. The figure 2 shows the schematic block diagram of the speed controlled system. A proportional integral controller is used in outer loop for speed regulation. For increasing the efficiency of the EVs PID controller is appropriate in place of PI it gives a fast response of the speed. As shown in figure 2, the speed error is the input of the speed controller block, that generate the electromagnetic torque, the speed error can be minimized by increasing or decreasing the electromagnetic torque in the machine. The speed controller block also generates the mutual flux linkages reference according the demand of the rotor speed. The electromagnetic torque reference and the mutual flux linkage reference synthesis the reference stator current, that can be used in a inner loop that is torque controlled of the drive system.